Mergestack

5.1 Pg 236

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Q1) $f: A \to B$

$C \subseteq A$

$f|C = f \cap (C \times B)$

In this proof, when we have $f: A \to B$ & we have $x \in A$ we can apply property of fction to get $f(x) \in B$

(a)

$f|C$ is a fction $C \to B$

$\forall c \in C \exists! b \in B (c, b) \in f|C$

Let $c \in C$ be arbitrary

$\exists! b \in B (c, b) \in f \cap (C \times B)$

Supp existence: Note that $c \in A$ as well

Let $b = f(c)$ so $b \in B$. so $(c, b) \in C \times B$ as $C \subseteq A$ so $c \in A$ so $(c, b) \in f$. so $(c, b) \in f \cap (C \times B)$

Uniqueness: $\forall b_1 \in B [(c, b_1) \in f|C \to b_1 = b]$

Supp $b_1 \in B$

& $(c, b_1) \in f|C$

$b_1 = b$

so $(c, b_1) \in f \wedge (c, b_1) \in C \times B$

but we have $(c, b) \in f$ & f is a fction so it must be that $b = b_1$.

To derive formula $\forall c \in C [f(c) = (f|C)(c)]$

~~$(f|C)(c)$~~

Supp $c \in C$

$\to$ Supp $f(c)$ $\to (f|C)(c)$

Let $b = f(c) \in B$ $f \cap (C \times B)(c)$

so $(c, b) \in f$ $(c, b) \in f \cap (C \times B)$

Clearly $c \in C$ & $b \in B$ so $(c, b) \in C \times B$ & we already know that $(c, b) \in f$

since we already proved it's a fction

$\leftarrow$ Supp $(c, b) \in f|C$. Clearly $(c, b) \in f$ as $f|C = f \cap (C \times B)$.

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(b) $g : C \to B$ $\quad g = f|C \iff g \subseteq f$

$\to$ Supp $g = f|C$ $\quad g \subseteq f$

Supp $(c,b) \in g$ be arbitrary $\quad (c,b) \in f$

As $g = f|C$ so $(c,b) \in f|C$ so $(c,b) \in f \cap (C \times B)$

so $(c,b) \in f$.

$\leftarrow$ Supp $g \subseteq f$ $\quad g = f|C$

Any arbitrary $(c,b) \in g$ iff $(c,b) \in C \times B \land (c,b) \in f$ ($\because g \subseteq f$)

iff $(c,b) \in f \cap (C \times B)$ iff $(c,b) \in f|C$

(c) $g = \{(x,y) \in \mathbb{Z} \times \mathbb{R} \mid y = \underbrace{2x+3}_{g(x)}\}$

$h = \{(x,y) \in \mathbb{R} \times \mathbb{R} \mid y = \underbrace{2x+3}_{h(x)}\}$

$g = h|\mathbb{Z}$

$\to$ Supp $(x,y) \in g$. $\quad (x,y) \in h|\mathbb{Z}$

so $(x,y) \in \mathbb{Z} \times \mathbb{R} \land y = 2x+3$ $\quad (x,y) \in h \cap (\mathbb{Z} \times \mathbb{R})$

to see that $(x,y) \in h$

as $\mathbb{Z} \subseteq \mathbb{R}$ so $x \in \mathbb{R}$ as well so $(x,y) \in \mathbb{R} \times \mathbb{R}$ as well

& $y = 2x+3$ as we already know.

$\leftarrow$ Supp $(x,y) \in h|\mathbb{Z}$ $\quad (x,y) \in g$

iff $(x,y) \in \mathbb{R} \times \mathbb{R} \land (x,y) \in \mathbb{Z} \times \mathbb{R} \land y = 2x+3$

clearly goal follows.

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(8) $f : A \to B$ $\quad \exists A' \subseteq A \,(g : A' \to B)$

$g \subseteq f$ $\quad g \subseteq B$

let $A' = \{a \mid (a,b) \in g\}$. Domain of g ①

let $a \in A'$ be arbitrary as ~~then~~ $(a,b) \in g$ & $g \subseteq f$

so $(a,b) \in f$ so $a \in A$ so $A' \subseteq A$.

To see that g is a function $\quad \forall a \in A' \exists! b \in B \,(a,b) \in g$

let $a \in A'$. $A' \subseteq A$ so $a \in A$ as well

~~let $b = f(a) \in B$ so $(a,b) \in f$~~

then by ① we defined there exist some $b \in B$ with $(a,b) \in g$

Uniqueness

Supp $(a,b_1) \in g \land (a,b_2) \in g$. $\quad b_1 = b_2$

so $a \in A'$, $b_1 \land b_2 \in B$ $\quad \because g \subseteq f$

as $A' \subseteq A$ so $a \in A$

so $(a,b_1) \land (a,b_2) \in f$ but f is a function so $b_1 = b_2$

(Q 9) $f : A \to B$

$B \neq \emptyset$ $\quad \exists g : A' \to B \,(f \subseteq g)$

$A \subseteq A'$

By analyzing an example we can see that we don't want to repeat the domain. $\because B \neq \emptyset$ so let $b_0 \in B$. but this be only element that we will map elements of A' so domain won't be repeated. Also g should contain f. So after analyzing an example $g = f \cup \{(A' \setminus A) \times \{b_0\}\}$ looks the right choice

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we will prove first that $g : A' \to B$

Existence: supp $a \in A'$. Then there is $b_0 \in B$ such that by definition of g, $(a, b_0) \in g$.

Uniqueness:

supp $(a, b_1) \in g \wedge (a, b_2) \in g$. $\quad b_1 = b_2$

: As $g = f \cup \{(A' \setminus A) \times \{b_0\}\}$. We consider 2 cases

Case 1) supp $(a, b_1) \in f$

Case 1.1) $(a, b_2) \in f$. then f is function so $b_1 = b_2$

Case 1.2) $(a, b_2) \in \{(A' \setminus A) \times \{b_0\}\}$ then $a \in A' \wedge a \notin A$ which is contradiction over $a \in A$ as $(a, b_1) \in f$. So this case is not possible.

Case 2) $(a, b_1) \in \{(A' \setminus A) \times \{b_0\}\}$. Then $b_1 = b_0$

if $(a, b_2) \in f$. then $a \in A$ which is contradiction over $a \notin A$. So this is not possible.

so $(a, b_2) \in \{(A' \setminus A) \times \{b_0\}\}$ so $b_2 = b_0$

so $b_1 = b_2 = b_0$.

Hence proved.

Finally to see that $f \subseteq g$

supp $(a, b) \in f$. then $\to (a, b) \in g$

as $g = f \cup \{(A' \setminus A) \times \{b_0\}\}$ so $(a, b) \in g$.

(Q10) $f : A \to B$, $g : A \to B$, $f \neq g$.

$f \triangle g$ is not a function

$= (f \cup g) \setminus (f \cap g)$ is not a function

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By translating logical symbols, we can see that as $f \neq g$ we can choose some $(a, b) \in A \times B$ such that $(a, b) \in f \wedge (a, b) \notin g$ $\vee$ $(a, b) \in g \wedge (a, b) \notin f$.

Case 1) $(a, b) \in f \wedge (a, b) \notin g$.

We will use theorem that $f = g$ iff $\forall a \in A\,(f(a) = g(a))$

As $f \neq g$. so we can choose some $a_0 \in A\,(f(a_0) \neq g(a_0))$

Since $f : A \to B$ so $(a_0, f(a_0)) \in f$. — and $(a_0, f(a_0)) \notin g$ as g is a function so $(a_0, f(a_0)) \in f \triangle g$

So similarly $(a_0, g(a_0)) \in g$ → similar argument

Since $f(a_0) \neq g(a_0)$ so $f \triangle g$ is not a function.

(Q11) we have already seen that it is equivalence in Ex: 4.8. we saw in Ex: 5 it's a function.

Uniqueness:

supp f & g are 2 functions on A that are also equivalence relations $\longrightarrow$ $f = g$, $\forall a \in A\,(f(a) = g(a))$

so $f : A \to A$, $g : A \to A$

let $a \in A$.

By reflexivity of f & g. $(a, a) \in f$ & $(a, a) \in g$

so $f(a) = a = g(a)$. Hence proved

(Q12)

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Date: ___ / ___ / ___ $f, g : A \to C$ MON – TUE – WED – THUR – FRI – SAT – SUN

Q12) (b) → Supp $f \cup g : A \cup B \to C$

$f|(A \cap B) = g|(A \cap B)$

~~$f|A \cap B \subseteq g|A \cap B$~~

~~$f|A \cap B \subseteq g|A \cap B \times C$~~

~~$\forall a \, (f|(A \cap B))(a) = g|(A \cap B)(a)$~~

~~Supp $(x, y) \in f|(A \cap B)$~~

~~iff $(x, y) \in f \cap (A \cap B) \times C$~~

~~so $x \in A \cap B$ so $x \in B$~~

~~As $g : B \to C$~~

~~so $\exists y_1 \in C \ (x, y_1) \in g$~~

~~If we can prove $y_1 = y$ then we are done~~

~~as $x \in A \cap B$ so $x \in A \cup B$~~

~~so $\exists y_2 \in C \ (x, y_2) \in f \cup g$~~

~~(case 1) $(x, y_2) \in f$ then $y = y_2$ (as f is function)~~

~~(case 2) $(x, y_2) \in g$ then $y_1 = y_2$~~

~~so $f = g$~~

~~we need to prove as functions $A \cap B \to C$~~

By existence that as both are 2 are functions

→ so let $a \in A \cap B$ be arbitrary

$\forall a \in A \cap B \ [(f|A \cap B)(a) = (g|A \cap B)(a)]$

Supp $a \in A \cap B$ then $a \in A$ and $a \in B$. As $f : A \to C$ & $g : B \to C$

$\exists c_1 \in C \ f(a) = c_1$ — ①

$\exists c_2 \in C \ g(a) = c_2$ — ②

As $(a, c_1) \in f$ so $(a, c_1) \in f \cup g$ similarly $(a, c_2) \in f \cup g$

but $f \cup g : A \cup B \to C$ so $c_1 = c_2$.

Note: The problem was that I was considering cases that $a \in A$

so $a \in A \cup B$. Then there are 2 cases: $a \in A$ & $a \in B$. But

we would also try $f \cup g$ as well. $a \in A \cap B$, so using any of

∴ not as powerful as using $f \cup g$ given $A \cup B$

$\forall a \in A \ \exists! c \in C \ (a, c) \in f \cup g$

[illegible] $(a, c) \in f \cup g$

← Supp $f|A \cap B = g|A \cap B$ $f \cup g : A \cup B \to C$

Supp $a \in A \cup B$ $\exists! c \in C \ (a, c) \in f \cup g$

(case 1) $a \in A$. As $f : A \to C$. Let $f(a) = c_1 \in C$ $\therefore (a, c_1) \in f$ so $(a, c_1) \in f \cup g$

(case 2) similar

Uniqueness:

Supp $(a, c_1) \in f \cup g$ & $(a, c_2) \in f \cup g$. There are 4 cases

The cases when $(a, c_1), (a, c_2) \in f$ & $(a, c_1), (a, c_2) \in g$ are straight. As f, g are functions so $c_1 = c_2$.

(case 3) $(a, c_1) \in f$ & $(a, c_2) \in g$.

so $a \in A$ & $a \in B$ so $a \in A \cap B$. ~~As $(f|A \cap B)$~~

so $(a, c_1) \in f|A \cap B = g|A \cap B$ so $(a, c_1) \in g|A \cap B$

so $(a, c_1) \in g$. As also $(a, c_2) \in g$ & g is a function

so $c_1 = c_2$.

Q13) $R \subseteq A \times B$, $S \subseteq B \times C$

$Ran(R) = Dom(S) = B$

$S \circ R : A \to C$

(a) $S : B \to C$

Supp $b \in B$ $\exists! c \in C \ (b, c) \in S$

Existence: As $B = Dom(S)$ so $b \in Dom(S)$ so $\exists c \in C \ (b, c) \in S$

Uniqueness: Supp $(b, c') \in S$. As $b \in B = Ran(R)$ so there exists some $a \in A \ (a, b) \in R$. As $(a, b) \in R$, $(b, c) \in S$ so $(a, c) \in S \circ R$. Similarly

$(a, c') \in S \circ R$ but $S \circ R$ is a function so $c = c'$.

S. [illegible] a function

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(c) $A=\{1,2,3\}, B=\{a,b\}, C=\{1,2\}$

$R=\{(1,1),(1,2),(2,1)\}$

$S=\{(1,2),(2,1)\}$

$S \circ R=\{(1,2),(2,2)\}$

Q13) Note: The uniqueness proof pattern is kind of proof by contradiction in which we contradict the fact that something is a function (e.g. f) so (a,b_1) & (a,b_2) contradicts along w. f so $b_1=b_2$ a must. Although, in scope of $\exists!$ we wanted that pair of 2 elements satisfying some property, they must be equal.

Q14) $f: A\to B$ $\quad S\subseteq B\times B$

$R=\{(x,y)\in A\times A \mid (f(x),f(y))\in S\}$

(a) S is reflexive $\longrightarrow$ R is reflexive

$\forall b\in B\,(b,b)\in S$ $\longrightarrow$ $\forall a\in A\,(a,a)\in R$

Supp $a\in A$ then $\longrightarrow$ $(a,a)\in R$

$\exists! b\in B\,(a,b)\in f$ $(f(a)=b)$ $\quad$ $(f(a),f(a))\in S$

$f(a)=b\in B$

As S is refl, $(b,b)\in S$ or $(f(a),f(a))\in S$

(b) S is symmetric $\longrightarrow$ R is symmetric

let $a_1,a_2\in R$ & $a_1 R a_2$ $\longrightarrow$ $a_2 R a_1$

$a_1 R a_2 \to (f(a_1),f(a_2))\in S$

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f is symmetric so $(f(a_2),f(a_1))$ using def of R, $a_2 R a_1$

(c) S is transitive $\longrightarrow$ R is transitive

Suppose $a_1,a_2,a_3\in A$ & $a_1 R a_2$ & $a_2 R a_3$ $\longrightarrow a_1 R a_3$

$f(a_1) S f(a_2)$ & $f(a_2) S f(a_3)$ by transitivity of S, $f(a_1) S f(a_3)$

So $a_1 R a_3$.

Q15) $f: A\to B$, $R\subseteq A\times A$

$S=\{(x,y)\in B\times B \mid \exists u,v\in A\,(f(u)=x \wedge f(v)=y \wedge (u,v)\in R)\}$

(a) Already solved in book. Not all elements of B [illegible]

(c) It will help to see that when supp $b_1 S b_2$ & $b_2 S b_3$ then $f(a_1)=b_1$ & $f(a_2)=b_2$ but for $b_2 S b_3$, there can be $f(a_2')=b_2$ (not need to be a_2 again)

Q16) A & B are sets $F=\{f \mid f: A\to B\}$

$R\subseteq B\times B$

$S=\{(f,g)\in F\times F \mid \forall x\in A\,((f(x),g(x))\in R)\}$

(a) Yes. Suppose $f\in F$ $\longrightarrow (f,f)\in S$

let $x\in A$. As $f: A\to B$ so $\exists! b\in B\,(f(x)=b)$. As R is refl., so $(b,b)\in R = (f(x),f(x))\in R$ so $(f,f)\in S$

(b) Yes. Suppose $(f,g)\in S$ then $\forall x\,((f(x),g(x))\in R$

Let $x \in A$ $\longrightarrow$ $(g, f) \in S$

As $x \in A$, so $(f(x), g(x)) \in R$ & R is symm, so $(g(x), f(x)) \in R$

so $(g, f) \in S$.

(c) Pre easy proof similar to symmetry.

(Q 17) $A \neq \phi$, $f : A \to A$

$\exists a \in A \, \forall x \in A \, (f(x) = a)$ —① $\quad \forall g : A \to A \, (f \circ g = f)$

Let $g : A \to A$ be arbitrary

$\to$ Supp $(a_1, a_2) \in f \circ g$. Then $(a_1, x) \in g \wedge (x, a_2) \in f$. By ①,
$\forall x \in A \, (f(x) = a_2)$ but also $f(x) = a_2$ so $a_2 = a_2$
so $\forall x \in A \, (f(x) = a_2)$. Since $a_1 \in A$ so $f(a_1) = a_2$ so $(a_1, a_2) \in f$.

$\leftarrow$ Supp $f(a_1) = a_2$ as $(a_1, a_2) \in f$. As $\forall x \in A \, (f(x) = a_2)$, we know $a_1 \in A$ so $f(a_1) = a_2$ so $a_2 = a_2 \wedge \forall x \in A \, (f(x) = a_2)$

Stuck: Do not forget that to prove that 2 functions are equal, use the relation $f = g$ iff $\forall x \, [f(x) = g(x)]$

So to prove this here: supp $g : A \to A$

Now prove $\forall x \in A \, (f \circ g(x) = f(x))$

Let $x \in A$: $f \circ g(x) = f(g(x))$ as ~~[illegible]~~ $g : A \to A$
$\exists! a_1 \in A \, (g(x) = a_1)$ so $f(g(x)) = f(a_1)$ & as
$f : A \to A$ $\exists a \in A \, (f(a_1) = a)$. By the given ①,
we can conclude $\forall y \in A \, (f(y) = a)$ so as $x \in A$
$f(x) = a$ as well so $f \circ g(x) = f(g(x)) = f(a_1) = a = f(x)$
so $f \circ g(x) = f(x)$

(b) $\forall g : A \to A \, (f \circ g = f)$ $\longrightarrow$ f is constant
$\exists a \in A \, \forall x \in A \, (f(x) = a)$

$\leftrightarrow \forall g : A \to A \, [\boxed{\forall x \in A \, (f \circ g(x) = f(x))}]$

$\forall g : A \to A \, [\forall x \in A \, (f(g(x)) = f(x))]$

Intuition & understanding

First $A \neq \phi$ so select $a_0 \in A$. An important thing is that f is a function, so select $f(a_0) = a$. Last important thing is that ~~[illegible]~~ to apply universal instantiation, we have to choose any g that is function (to use this given). Our only choice is to define $\forall x \in A \, (g(x) = a_0)$ as a_0 is fully defined here

Note that this does make g a constant function but to apply universal instantiation, (which needs for all $g : A \to A$), we just need one g that is function.

$K \subseteq \mathbb{R}$ are not same

(Q18) $F = \{f \mid f : \mathbb{R} \to \mathbb{R}\}$

$R = \{(f, g) \in F \times F \mid \exists a \in \mathbb{R} \, \forall x > a \, (f(x) = g(x))\}$

(a) $f : \mathbb{R} \to \mathbb{R}$ $\quad f(x) = |x|$
$g : \mathbb{R} \to \mathbb{R}$ $\quad g(x) = x$ $\qquad (f, g) \in R$

f & g are already function. $\qquad \exists a \in \mathbb{R} \, \forall x > a \, (f(x) = \ldots$

Let $a = 0$

then $\forall x > 0$ $\quad f(x) = |x| = x = g(x)$ $\qquad$ ~~$\forall f$~~ $\forall g \, [(f, g) \in R \to (g, f) \in R$

(b) ~~Reflexivity~~ Symmetry $\qquad$ ~~$\forall (f, g) \in R \; ((g, f) \in R)$~~)

Let $(f, g) \in R$ then $\exists a \in \mathbb{R} \, \forall x > a \, (f(x) = g(x))$

which is same as $\exists a \in \mathbb{R} \, \forall x > a \, (g(x) = f(x))$ so $(g, f) \in R$

~~Symmetry~~ Reflexivity $\forall f \in F[(f,f) \in R]$

let $f \in F$. $\exists a \in \mathbb{R}\ \forall x > a\ (f(x) = f(x))$

Clearly for all x, $f(x) = f(x)$ as f is a function.
So it must be true for some $x > a$

Transitivity:

$\forall f \forall g \forall h [fRg \wedge gRh \to fRh]$

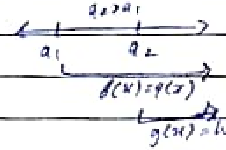
let $f, g, h \in F$
& supp $fRg \wedge gRh$ $\longrightarrow$ fRh

fRg means $a_1 \in \mathbb{R}\ \forall x > a_1\ (f(x) = g(x))$

gRh means $a_2 \in \mathbb{R}\ \forall x > a_2\ (g(x) = h(x))$

we have to prove $\exists a \in \mathbb{R}\ \forall x > a\ (f(x) = h(x))$

We can find it by looking at number line. If $a_1 = a_2$ then $a = a_1 = a_2$ works. If $a_1 > a_2$ then $a = a_1$ works.
If $a_2 > a_1$ then $a = a_2$ works.



so for $x > a_2$ both $f(x) = g(x)$ & $g(x) = h(x)$ will hold. Similar for other 2 cases

The reason we arrived at cases is that if we look at goal & the givens. We have to prove $f(x) = h(x)$ & only way that looks is using $f(x) = g(x)$ & $g(x) = h(x)$ so there should be some a for which both of them holds true & we can conclude $f(x) = g(x) = h(x)$.
That a might be different in case $a_1 < a_2$ or $a_1 > a_2$ or $a_1 = a_2$.

Case 1) $a_1 = a_2$. Then let $a = a_1$. Clearly $\forall x > a_1$ $f(x) = g(x) = h(x)$ so fRh

Case 2) $a_1 < a_2$. then let $a = a_2$. let $x > a_2$ be arbitrary. As $x > a_2$ so $f(x) = h(x)$. also $x > a_2 > a_1$, so $x > a_1$, so $g(x) = h(x)$ & so $f(x) = h(x)$ so fRh

Case 3) $a_1 > a_2$. then let $a = a_1$. let $x > a_1$ be arbitrary. As $x > a_1$ so $f(x) = g(x)$. Also $x > a_1 > a_2$ so $x > a_2$ so $g(x) = h(x)$. therefore $f(x) = h(x)$ & thus fRh

Q19)(a) We just need to find one combination of a & c that works. One combination of a & c already implies infinite combinations as example for any $b > a$, the same c would work.

$F = \{f \mid f: \mathbb{Z}^+ \to \mathbb{R}\}$

for any $g \in F$, $O(g) = \{f \in F \mid \exists a \in \mathbb{Z}^+\ \exists c \in \mathbb{R}^+\ \forall x > a\ (|f(x)| \leq c|g(x)|)\}$

(a) $f(x) = 7x + 3 : \mathbb{Z}^+ \to \mathbb{R}$
$g(x) = x^2 : \mathbb{Z}^+ \to \mathbb{R}$ prove that $f \in O(g)$

$= \exists a \in \mathbb{Z}^+\ \exists c \in \mathbb{R}^+\ \forall x > a$.

let $a = 3$ & $c = 8$

Then $|f(x)| = f(x) = 7x + 3 \leq 7x + x = 8x \leq 8x^2$
$= c|g(x)|$.

Basically we needed to show $\forall x > 3\ [|7x + 3| \leq 8|x^2|]$
We can choose any a & c that works but choose one that is easiest to prove. For a different combination, we might have to go for proof by contraposition or contradiction.

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(b) There are 2 ways. Assume $\exists a \in R^+ \exists c \in R^+ \forall x > a [x^2 \leq c(7x+3)]$ or simply show the negation that is $\forall a \in R^+ \forall c \in R^+ \exists x > a (|g(x)| > c|f(x)|)$

let us do second

let $a \in R^+$ & $c \in R^+$ $\exists x > a [x^2 > c(7x+3)]$

let $x_0 = ...$ such that $x_0 > a$ & $x_0 > 10c$ $x_0^2 > c(7x+3x)$

Since $x_0 > 10c$ so $x_0^2 > 10cx = c(7x+3x)$

also $(7x+3x) \geq c(7x+3)$.

so we found an $x > a$ that prove the negation of statement

(c) $f_1 \in O(g)$

$f_2 \in O(g)$

$s, t \in R$

$f: R^+ \to R$ $f(x) = s f_1(x) + t f_2(x)$

$\to f \in O(g)$

$= \exists a \in R^+ \exists c \in R^+ \forall x > a [|f(x)| \leq c|g(x)|]$ — ①

$f_1 \in O(g)$

$\exists a_1 \in R^+ \exists c_1 \in R^+ \forall x > a_1 (|f_1(x)| \leq c_1|g(x)|)$ ②

$f_2 \in O(g)$

$= \exists a_2 \in R^+ \exists c_2 \in R^+ \forall x > a_2 (|f_2(x)| \leq c_2|g(x)|)$ ③

To prove the goal, we need to show that $f(x) = |s f_1(x) + t f_2(x)| \leq c|g(x)|$. We have to find a & c that satisfies this property for all $x > a$. Note we can use $|f_1(x)| \leq c_1|g(x)|$ if $x > a_1$ & similarly

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for a_2 so $x > a_1$ & $x > a_2$ should be choosed. To choose c note that $|s f_1(x) + t f_2(x)| = |s f_1(x)| + |t f_2(x)|$

Already $|f_1(x)| \leq c_1|g(x)|$ & $|f_2(x)| \leq c_2|g(x)|$

so this hints toward $c_1 + c_2$ as ~~no~~ $c|g(x)|$ become $c_1|g(x)| + c_2|g(x)|$. Now only problem of s & t remains so we can multiply with s & t also but their absolute bec they can be negative so $c_1|s| + c_2|t|$ works. Finally s & t can be 0, so we just add 1 to make sure $c \in R^+$ $c = c_1|s| + c_2|t| + 1$

The proof is simple now.

let $a = \max(a_1, a_2)$ ($\because$ Now $a > a_1$ & $a > a_2$)

let $c = c_1|s| + c_2|t| + 1$

let $x > a$ be arbitrary then also $x > a_1$ & $x > a_2$. (Now we can use ② & ③)

then $|f(x)| = |s f_1(x) + t f_2(x)| = |s f_1(x)| + |t f_2(x)|$

$\leq c_1|s||g(x)| + c_2|t||g(x)| = (c_1|s| + c_2|t|)|g(x)|$

$\leq c|g(x)|$. ($\because$ as we didnt write +1 so $\leq$ should be used)

$\because$ We used as $|f_1(x)| \leq c_1|g(x)|$

multiply by $|s| \to |s||f_1(x)| \leq c_1|s||g(x)|$

(Q20) $g: A \to B$

$R = \{(x, y) \in A \times A \mid g(x) = g(y)\}$

R is equivalence

Reflexivity $\forall x \in A (x, x) \in R$

Supp $x \in A$ then clearly $(x, x) \in A \times A$ & $g(x) = g(x)$

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so $(x,x) \in R$

Symmetry $\forall x \forall y \in A \, (xRy \to yRx)$

Supp $x \in A \wedge y \in A \wedge xRy$ $\to yRx$

xRy means $g(x) = g(y)$ which is same as $g(y) = g(x)$

so yRx.

Transitivity $\forall x \in A \, \forall y \in A \, \forall z \in A \, [(xRy \wedge yRz) \to xRz]$

Supp $x \in A, y \in A, z \in A$

Supp $xRy \wedge yRz$ $\to xRz$

which means $g(x) = g(y) \wedge g(y) = g(z)$ so $g(x) = g(z)$

so xRz.

---

b) R is equivalence on A

$g: A \to A/R \mid g(x) = [x]_R$

Prove that $R = \{(x,y) \in A \times A \mid g(x) = g(y)\}$

note that R has simply those pairs (x,y) where $[x]_R = [y]_R$

$\forall x \in A \, \forall y \in A \, [(x,y) \in R \to (x,y) \in \{(x,y) \in A \times A \mid g(x) = g(y)\}]$

Supp $(x,y) \in R$

By symmetry of R, yRx — ①

so $y \in [x]_R$: by def of $[x]_R$ & ①

so $[x]_R = [y]_R$: Lemma 4.5.5.

so $g(x) = g(y)$ where $(x,y) \in \{(x,y) \in A \times A \mid g(x) = g(y)\}$

This is a series of equivalent statements that imply both sides.

note that R here was like saying $R = \bigcup_{X \in A/R} X \times X$

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21) Note: The compatibility relation (ex 24, sec 4.5) tells us that R is compatible with S if any 2 elements are related by R iff other elements of their same equivalence class are also related by R so R respects the tendencies of S.

The compatibility of functions that any 2 elements who have same class must have same function output.

So f collapses every class of A/R to a single output

(a) $f: A \to B$

R is equivalence relation on A

f is compatible with R = $\forall x \in A \, \forall y \in A \, (xRy \to f(x) = f(y))$

Existence: Let $h = \{([x], y) \in A/R \times B \mid \exists z \in [x] \, (f(z) = y)\}$

Note: This can be reached by analyzing a small example. Also its same as saying $\forall z \in [x] \, (f(z) = y)$ bcz f makes sure every elements in $[x]$ has same output in f. But mathematicians use $\exists$ to prevent empty sets & easy proofs

h is a Function

$\forall [x] \in A/R \, \exists! y \in B \, (h([x]) = y)$

Let $[x] \in A/R$ $\to h([x]) = y = \exists z \in [x] \, (f(z) = y)$

Existence: Let $y = f(x) \in B$ (∵ use the same x)

clearly $x \in [x]$ & $f(x) = y$ $f = z$

Uniqueness of y: Let $y_1 \in B \, (h([x]) = y_1)$ — ①

Let $y_2 \in B \, (h([x]) = y_2)$ — ②

① $\to x_1 \in [x] \, (f(x_1) = y_1)$ ∵ By def of h

(ii): $\exists x_2 \in [x] \, (f(x_2) = y_2)$
∴ $x_1 R x \land x_2 R x$; By def of $[x]$
So $f(x_1) = f(x_2) = f(x)$; By compatibility of f
So $y_1 = y_2$; $f(x_1) = y_1 \land f(x_2) = y_2$

Uniqueness of h :
Supp $h_1 : A/R \to B$ — (i)
$h_2 : A/R \to B$ — (ii)
→ $h_1 = h_2$
→ $\forall [x] \in A/R \, [h_1([x]) = h_2([x])]$
Let $[x] \in A/R$ → $h_1([x]) = h_2([x])$
(i): $h_1 = \{([x], y_1) \in A/R \times B \mid \exists x_1 \in [x] \, (f(x_1) = y_1)\}$
(ii): $h_2 = \{([x], y_2) \in A/R \times B \mid \exists x_2 \in [x] \, (f(x_2) = y_2)\}$
; By def of h
∴ $x_1 R x \land x_2 R x$; By def of $[x]$
∴ $f(x) = f(x_1) = f(x_2)$; By compatibility of f
$f(x) = y_1 = y_2$; $f(x_1) = y_1 \land f(x_2) = y_2$
∴ $h_1([x]) = y_1 = y_2 = h_2([x])$
Hence Proved!

(a) Finally → $\forall x \in A \, [h([x]) = f(x)]$
Supp $x \in A$ → $h([x]) = f(x)$
Clearly $x \in [x]$
As h is a function from $A/R \to B$
$\exists! y \in B \, [\exists x_1 \in [x] \, (f(x_1) = y)]$
As $x \in [x]$ so $f(x) = y = h([x])$

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(b) $h : A/R \to B$
$\forall x \in A \, (h([x]) = f(x))$ — (1) $\forall x \in A \, \forall y \in A \, (xRy \to f(x) = f(y))$
Let $x \in A \land y \in A \land xRy$
(1): $h([x]) = f(x)$
(1): $h([y]) = f(y)$
But since xRy, so $y \in [x]$ so $[y] = [x]$; Lemma 4.5.5
so $h([x]) = f(x) = h([y]) = f(y)$
so $f(x) = f(y)$
Hence Proved!

22) By making classes & analyzing, we can they turn out to be:
$\{0, 5, 10, 15 \ldots\}$
$\{1, 6, 11, 16, 21, 26 \ldots\}$
$\{2, 7, 12, 17, 22, 27 \ldots\}$
$\{3, 8, 13, 18, 23, 28 \ldots\}$
$\{4, 9, 14, 19, 24, 29 \ldots\}$

(a) This can be simply proved by defining F & showing that F is compatible with R. The rest of proof will follow from Exercise 21.
The proof that $F = \{(x, [x^2]) \in N \times N/R \mid f(x) = [x^2]\}$ is compatible with R is easy. Supp $x \in N, y \in N$ & xRy which means $[x] = [y]$. Then showing $[x^2] = [y^2]$ is another proof given on whiteboard. Compatibility will follow from that proof.

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Another comprehensive way is to define h

$h = \{([x],[y]) \in N/R \times N/R \mid \exists x_1 \in [x]([x_1^2] = [y])\}$

This shows h is a function & the uniqueness of h

h can also be defined as (using F)

$h = \{([x],[y]) \in N/R \times N/R \mid \exists x_1 \in [x](f(x_1) = [y])\}$

In both cases, whole proof is similar.

(b) This is a proof by contradiction

Assume h exists

$h : N/R \to N/R (\forall x \in N (h([x]) = [2^x])$

Then $h([0]) = [2^0] = [1]$

& $h([5]) = [2^5] = [32]$

$[0] = [5]$ but $[1] \neq [32]$. This contradicts the fact that h is a function. ($[0] = [5]$ so this is same input but we got 2 different outputs $[1] \neq [32]$)

Some Notes about $O(x) \to Q19$

The big-O is any upper bound (e.g. $7x+3 \in O(x)$. Also $7x+3 \in O(x^2)$) In Algorithm analysis we use Θ which shows that it is a tight bound (both upper & lower bound). This shows exact growth rate. like $O(x)$ mean upper bound, Ω mean lower bound. ie

$f \in \Omega(g)$ iff $\exists a \in Z^+ \; \exists c \in R^+ \; \forall x > a [|f(x)| \geq c|g(x)|]$

Now $f \in \Theta(g)$ iff $f \in O(g) \wedge f \in \Omega(g)$

iff $f \in O(g) \wedge g \in O(f)$

h is ~~[illegible]~~ Existence $\exists! [y] \in N/R \, (h([x]) = [y])$

let $[x] \in N/R$ $\rightarrow \exists x_1 \in [x] \, (f(x_1) = [y])$

$y = x^2$

$\therefore [y] = [x^2]$

Uniqueness : $[y_1] \in N/R$, $[y_2] \in N/R$ $[y_1] = [y_2]$

$\exists x_1 \in [x] \, (f(x_1) = [y_1])$

$\exists x_2 \in [x] \, (f(x_2) = [y_2])$ — ①

$[x_1] = [x] = [x_2]$ $(\because x_1 \in [x]$ & $x_2 \in [x])$

$\therefore$ as $[y_1] = [x_1^2]$ & $[y_2] = [x_2^2]$ — ②

① — $[x_1^2] = [x_2^2]$ — ③ (① & ②)

$[y_1] = [y_2]$

To see $[x_1] = [x_2] \rightarrow [x_1^2] = [x_2^2]$

$\exists k \in \mathbb{Z} \, (5k = x_1 - x_2) \rightarrow \exists m \in \mathbb{Z} \, (5m = x_1^2 - x_2^2)$

Note $x_1^2 - x_2^2 = (x_1 - x_2)(x_1 + x_2)$, $5k(x_1 + x_2)$

so let $m = k(x_1 + x_2)$

$5m = 5k(x_1 + x_2)$

$5m = (x_1 - x_2)(x_1 + x_2)$

$5m = x_1^2 - x_2^2$

Proved !

Reason : $\forall x \, \forall y \, (x = y \rightarrow [x] = [y])$

but reverse is not true

To define with F

$\exists x_1 \in [x] \, (f(x_1) = [y])$

This holds for $\forall x$ in $[x]$

$\{0, 5, 10, 15, 20, 25, 30, 35 \ldots\}$

$\{1, 6, 11, 16, 21, 26, 31, 36 \ldots\}$

$\{2, 7, 12, 17, 22, 27, 32, 37 \ldots\}$

$\{3, 8, 13, 18, 23, 28, 33, 38 \ldots\}$

$\{4, 9, 14, 19, 24, 29, 34, 39 \ldots\}$

$N/R = N/\equiv_5 = F =$

$h : \{([x], [y]) \in N/R \times N/R \mid \exists x_1 \in [x] \, (F(x_1) = [y])\}$

$F : \{(x, [x^2]) \mid f(x) = [x^2]\}$ $\in N \times N/R$

Uniqueness of h

$h_1 : N/R \rightarrow N/R$ — ①

$h_2 : N/R \rightarrow N/R$ — ②

$h_1 = h_2$

$\forall [x] \in N/R \, (h_1([x]) = h_2([x]))$

$h_1([x]) = h_2([x])$

suppose $[x] \in N/R$

① $\rightarrow \exists! [y_1] \in N/R \, (\exists x_1 \in [x] \, ([x_1^2] = [y_1]))$

② $\rightarrow \exists! [y_2] \in N/R \, (\exists x_2 \in [x] \, ([x_2^2] = [y_2]))$

Proof similar to proof that <u>h is a function</u>